AN2DL - Second Challenge Report
Budino Lovers

Carlo Bonandrini, Simone Decio, Jonel Relucio, Pietro Filippo Schgor

275682, 308376, 259768, 258727

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1 Introduction

In this challenge a collection of **images** was given. Each image comes from a low-magnification WSI *(Whole Slide Imaging)* of **human tissue**. The objective of this task is to perform *classification* by assigning each image to one of **four possible classes** corresponding to **molecular subtype** identifying a **potential disease within the tissue**, using models based on *Convolutional Neural Networks* architectures, **optimizing for the resulting F1-score**.

2 Problem Analysis

A *training dataset* of 691 labeled samples and a *test dataset* of 477 unlabeled samples were provided. Each sample is composed of a pair of features:

- *Image*: RGB image of variable size (around 1024x1024) of a tissue.
- *Mask*: binary mask, **identifying the regions most likely to contain the diseased tissue** in the associated image.

An initial inspection of the training dataset revealed two types of corrupted images. The first subset consisted of images of **Shrek** [1], cropped to resemble tissue, while the second subset contained images affected by bright green stains. As these images were detrimental to the task, they were discarded, resulting in a clean training dataset.

To reduce the impact of irrelevant background, a **patch-based cropping strategy** was applied, dividing each image into 128×128 patches with a stride of half the patch size. Only patches where the corresponding mask covered at least 70% of the area were retained. This approach mitigates background imbalance while preserving fine-grained details by avoiding downscaling and ensures that the model focuses on the most informative regions. To prevent information loss, at least one patch per image was always selected, choosing the patch with the highest overlap with positive mask areas when none met the threshold.

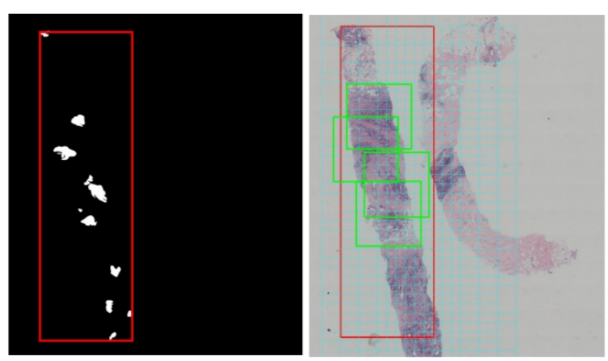

Figure 1: Left image shows the mask, right image displays the cropped patches.

3 Method

The model was optimized using a weighted **cross-entropy** loss, where class weights were defined as inversely proportional to the number of samples in each patch label. This weighting scheme mitigates class imbalance by assigning greater importance to underrepresented categories.

The training dataset was first split into training and validation set with a 85-15 split, ensuring that patches from the same image were not mixed between the sets. Subsequently, the patches are fed into a **pre-trained** model. The network layers are then **fine-tuned**, starting from the classifier and moving back to the earlier layers.

4 Experiments

Given the small size of the training dataset and the corresponding patches, a **transfer learning** approach was adopted. Several backbone models were evaluated, as summarized in Table 1. Ultimately, **ConvNeXt-Tiny** [6], pre-trained on *ImageNet*, was chosen. ConvNeXt modernizes the traditional CNN architecture by integrating key design principles from **Vision Transformers** [4] (layer normalization, GELU activations, and inverted bottleneck blocks) while retaining convolutional inductive biases. The Tiny variant was selected to balance model capacity with the risk of overfitting. Extensive data augmentation was applied, including:

- **Geometric transformations:** Random rotations, and horizontal/vertical flips.
- **Color jittering:** Adjustments to brightness, contrast, saturation, and hue to account for staining variations.
- **Regularization:** **MixUp** and **CutMix** [11], to encourage learning smoother decision boundaries.
- **Macenko color normalization:** a robust technique for color normalization of histology slides, reducing variability and improving consistency for quantitative analysis. [7]

4.1 Training Strategy

Warmup: The backbone of the network was frozen, and only the classification head, a linear layer mapping 768 features to four classes, was trained with a learning rate of 1×10^{-3}.

Fine-tuning: All blocks apart from the first two (`layer4`) and the classifier were unfrozen and trained using a reduced learning rate of 1×10^{-4} and a cosine annealing scheduler.

Test-Time Augmentation (TTA): During inference, TTA was applied using horizontal/vertical flips, up to 90° random rotations and color jittering. The final predictions for each image were obtained by averaging the softmax probability vectors from all patches and selecting the class with the maximum mean probability.

The AdamW optimizer was used with weight decay 1×10^{-2}. Early stopping with a patience of 100 epochs was applied on the validation macro F1-score.

5 Results

The best model achieved an F1-score of **0.4269** on the validation set and **0.4300** on the test set, using hyperparameters shown in Table 2. A more detailed analysis is needed to understand the sources of errors and mispredictions, guiding further enhancements to the model's overall performance.

5.1 Analysis

Examination of the per-class predictions reveals that, while Luminal B was often correctly predicted with high confidence, several high-confidence errors occurred for HER2(+) and Luminal A, suggesting that the model tends to be overconfident.

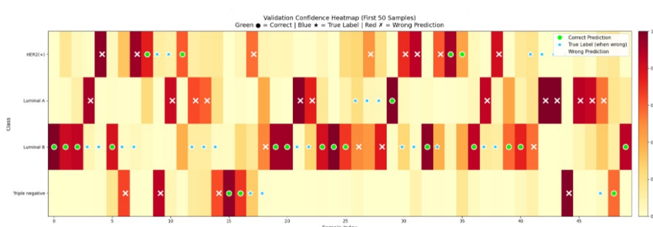

Figure 2: Confidence heatmap *(Val samples)*

For this reason, **self-training** or **pseudo-labeling** was not used. High-confidence errors would produce noisy labels, reinforcing mistakes and amplifying class-specific bias. With a limited dataset and no reliable uncertainty calibration, this approach was deemed unsafe and likely to harm generalization.

The bar chart details the per-class performance (Precision, Recall, F1-Score) for the four breast cancer subtypes. The model exhibits suboptimal performance across all categories,

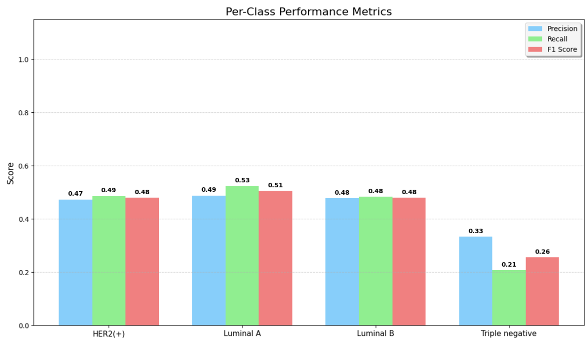

Figure 3: Precision, recall, F1-score per class

Model performance is severely bottlenecked by the Triple Negative subtype; its extremely low recall (0.21) remains the primary driver behind the overall score decrease, even when using weighted cross-entropy to counter class imbalance.

6 Discussion

To address these results, several alternative strategies were explored but did not yield performance improvements. These included a **mask-guided spatial attention** mechanism [8], where a spatial attention layer was applied to the first ConvNeXt feature map conditioned on given binary mask, and a **channel-attention** module inserted between the penultimate and final layers [9]. However, learned attention presents the evident problem of treating input patch edges as highly significant.

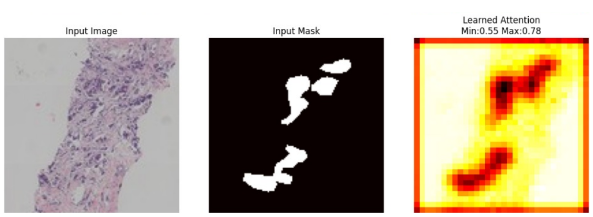

Figure 4: Learned attention *(white = 0, black = 1)*

Alternatively, the **binary mask** was incorporated either **as a fourth input channel** or through element-wise multiplication with the image, based on the idea that emphasizing regions of interest improves feature learning.

Multi-view data augmentation and training on multiple patch sizes were investigated to increase robustness to scale and contextual variation, while **K-fold cross-validation** was tested to improve generalization; although slightly beneficial, it proved computationally expensive. A **GAN** [2] [5] model for image generation was trained to address data scarcity, but the synthetic samples did not improve downstream performance.

Contrastive learning [3] was also attempted to enforce image similarity constraints, but training was slow and the model failed to learn meaningful representations, resulting in no observable gains. 5

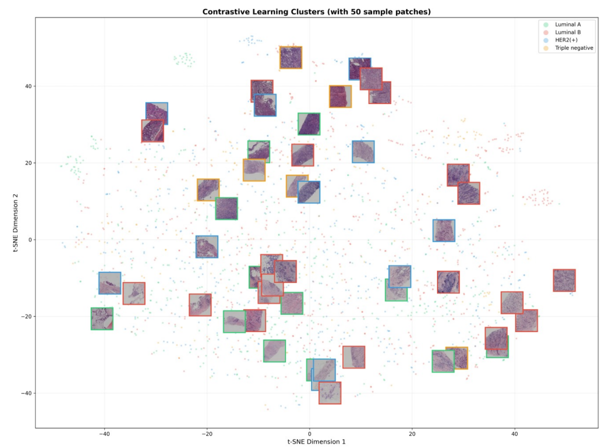

Figure 5: Learned Embedding Neighborhoods

7 Conclusions

With a final F1 score of 0.4300, the model significantly outperforms the random baseline (0.2932), validating its ability to learn morphological patterns. Nevertheless, it falls short of the competition's top performance (0.46+).
There is definitely space for future improvements, which may include:

- **Advanced Model Architectures:** Transitioning from CNNs to state-of-the-art architectures, such as Vision Transformers (ViT) to better capture global context.
- **Model Ensembling:** Implementing ensemble strategies,to reduce variance and improve prediction stability. *(**Latest result**: 30min before the deadline, the best F1-score of 0.4500 was reached with an ensemble of six models!)*
- **Intelligent Patch Extraction:** Refining the patch generation pipeline to prioritize tissues with high cellular density.
- **Advanced Synthetic Data Generation:** Investigating PixCell [10], a diffusion-based foundation model for histopathology synthesis enabling mask-conditioned generation for precise control over cell layout and tissue morphology during dataset expansion.

8 Appendix

Table 1: F1-scores of different models

Model	Validation F1	Test F1
Random Classifier	—	0.2932
SimpleCNN	0.3081	0.1612
EfficientNet-B0	0.3194	0.3535
EfficientNet-B1	0.3224	0.3212
EfficientNet-B2	0.3045	0.3200
ResNet18	0.4485	0.4162
ResNet50	0.3856	0.3803
ConvNeXt-Tiny	0.4269	0.4300
ConvNeXt-Small	0.4558	0.4171
ConvNeXt-Base	0.4039	0.4011
Ensembled Models		
6 ConvNeXt-Tiny	—	0.4500

GitHub: *https://github.com/jonelrelucio/AN2DL-2025*

Table 2: Training Hyperparameters

Hyperparameter	Value
Image Size	128 x 128
Batch Size	64
Rotation (degrees)	180
Horizontal Flip Probability	0.5
Vertical Flip Probability	0.5
Color Jitter - Brightness	0.1
Color Jitter - Contrast	0.1
Color Jitter - Saturation	0.1
Color Jitter - Hue	0.05
MixUp / CutMix Alpha	1.0
MixUp / CutMix Probability	0.5
Learning Rate (Warmup)	1×10^{-3}
Learning Rate (Finetune)	1×10^{-4}
Weight Decay (Warmup)	1×10^{-2}
Weight Decay (Finetune)	1×10^{-2}
Cosine Annealing T_{max}	300

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